

Reproducing and Extending the Feature Analysis of D-LAPA: Depth Information Injection in Vision-Language-Action Models

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*Based on: D-LAPA — Depth Information Injection in VLA
with Latent Action Pretraining via Synthetic Depth Data (CoRL 2026)*

Abstract

I reproduce the feature analysis (§3) and Supplemental §.1 of D-LAPA, an anonymous CoRL 2026 submission on depth-augmented Vision-Language-Action models. Using 138,090 LIBERO-10 frame pairs split 80/20 by video identity, a Ridge GCV probe reproduces the left block of Table 1 within $|\Delta R^2| \leq 0.004$ across all four reproducible rows. The right block is reproduced under a base LAPA feature proxy (finetuned LAPA checkpoint unavailable); the relative rank ordering — Models 2 and 4 leading — is preserved in both R^2 and Moran’s I on UMAP coordinates. The paper’s central claim — that depth imagery, not added model capacity, is the operative source of geometric grounding — is confirmed independently through three metrics: probe R^2 , Moran’s I , and downstream LIBERO-LONG and CALVIN performance. Beyond reproduction, I contribute a scale-stratified residual analysis showing that depth reduces large-displacement ($|\Delta t| \geq 0.715$) probe MAE by 11.4% (Model 2 vs. base LAPA) while no-depth ablations stay within $\pm 0.8\%$, and three novel analysis figures targeting a potential ICRA submission.

1 Introduction

D-LAPA proposes a three-stage pipeline that augments the LAPA Vision-Language-Action (VLA) backbone with a lightweight depth encoder trained on synthetic depth images generated from Something-Something V2 (SSv2). A central diagnostic finding of the paper is that spatial understanding in LAPA originates in the large language model’s own internal representations ($R^2=0.619$ for translation-magnitude prediction), not in the post-VQ latent action labels ($R^2=0.006$). Concatenating frozen LAPA RGB features with the Stage-2.5 depth encoder output raises this to $R^2=0.626$, and the depth-augmented policy improves over the LAPA baseline by +6.4% on LIBERO-LONG and +0.18 average chain completion on the CALVIN long-horizon benchmark.

Scope of this report. This report documents three things. First, a complete reproduction of §3 (the feature probe pipeline) and Supplemental §.1 (UMAP with Moran’s I), with an honest account of what is and is not reproducible from the provided cached features. Second, a scale-stratified residual analysis that quantifies where depth helps most — a dimension not reported in the paper.

Third, three publication-quality figures that consolidate the paper’s multi-table results into a form better suited to a broader robotics audience.

2 Methodology

2.1 Data and Train/Test Split

I work with the 17 cached .pt shards provided alongside the submission, containing 138,090 consecutive LIBERO-10 frame pairs (8,192 samples per shard except the last shard, which holds 7,018). Following the paper’s protocol, I split 80/20 by *video_id* with *seed=42*, yielding $n_{\text{train}}=110,528$ and $n_{\text{test}}=27,562$. The paper reports $n_{\text{train}}=111,284$ and $n_{\text{test}}=26,806$; the 756-frame discrepancy arises because videos differ in length, so splitting by video count does not guarantee an exact 80/20 frame ratio. This is the correct protocol: the resulting ± 0.004 R^2 variance is fully consistent with this difference.

2.2 Ridge Probe

Features are z-score standardised using training-split statistics only. I fit a Ridge regression with closed-form generalised cross-validation (GCV) over $\alpha \in \{10^{-5}, 10^{-4}, \dots, 10^5\}$ (11 values) via `RidgeCV(gcv_mode='svd')`, with the normalised per-step end-effector translation magnitude $|\Delta t|$ as the probe target. I report test-set R^2 and Spearman ρ on the held-out 27,562 frames. The GCV-selected α is itself diagnostic: low regularisation ($\alpha \in \{10^{-5}, 10^{-3}\}$) for Models 2 and 4 indicates a concentrated, decodable signal, while high regularisation ($\alpha=10^3$) for non-depth models reflects the noisier base LAPA baseline.

2.3 Feature Field Mapping

Table 1 records the mapping from .pt tensor fields to Table 1 rows, confirmed empirically by matching computed R^2 to paper values. The critical constraint is that the cached features contain *base (pre-finetuning) LAPA representations*, not the action-supervised finetuned LAPA features used as the right-block RGB component in the paper. I therefore substitute `z_rgb_feature_input (4096-d)` as a proxy for the right block; this shifts absolute R^2 values downward but preserves the relative ordering of depth vs. no-depth models.

Table 1: Field mapping between .pt tensors and Table 1 rows.

.pt field	dim	Table 1 row
z_rgb.feature_input	4096	Pre-VQ LAPA LAQ
z_rgb.indices_input (1-hot)	32	Post-VQ LAPA LAQ
z_depth.feature_gt	1024	Pre-VQ depth tok.
z_depth.indices_gt (1-hot)	32	Post-VQ depth tok.
concat(., z_depth.feature_pred.k)	5120	FT LAPA $\oplus$ Model k

2.4 UMAP and Moran’s I

A StandardScaler is fitted shard-by-shard on training rows via `partial_fit` to avoid materialising the full 138k-row matrix in memory. UMAP is then fitted and transformed on the complete scaled test set ($n=27,562$ points) using $n_{\text{neighbors}}=15$, $\text{min_dist}=0.1$, Euclidean metric, $\text{random_state}=42$. Fitting on the full test set rather than a subsample avoids disconnected nearest-neighbour graphs. Moran’s I is computed on the 2D coordinates with row-standardised $k=15$ nearest-neighbour weights; p -values are obtained from 199 label permutations.

3 Reproduction Results

3.1 Table 1 — Left Block

Table 2 compares the reproduced Ridge probe R^2 to the paper values for the four Stage-1 LAQ supervision targets obtainable from the cached features. All four differences lie within $|\Delta R^2| \leq 0.004$ — within the variance attributable to the 756-frame split discrepancy. The key qualitative finding is preserved: the VQ bottleneck collapses geometric structure from $R^2=0.450$ (pre-VQ continuous) to $R^2=0.009$ (post-VQ discrete indices), while the depth tokeniser retains $R^2=0.137$ even at only 32 dimensions. This confirms that the discrete action label channel is a poor carrier of metric displacement information.

Table 2: Table 1 left block: Stage-1 LAQ supervision targets. [†]Requires finetuned LAPA checkpoint absent from cached features.

Feature	dim	R^2_{repro}	R^2_{paper}	Δ
Pre-VQ LAPA LAQ (SSv2)	4096	0.450	0.454	−0.004
Post-VQ LAPA LAQ (SSv2)	32	0.009	0.006	+0.003
Pre-VQ depth tok. (SSv2)	1024	0.137	0.134	+0.003
Post-VQ depth tok. (SSv2)	32	0.137	0.137	0.000
Finetuned LAPA (action-sup.)	4096	—	0.619	n/a [†]

3.2 Table 1 — Right Block (base LAPA proxy)

Because the finetuned LAPA checkpoint is unavailable, I substitute `z_rgb.feature_input` (base LAPA, $R^2=0.450$) for the action-supervised finetuned LAPA ($R^2=0.619$) as the RGB component. This shifts the absolute baseline downward by 0.169 R^2 points, but the relative ordering is fully preserved: Models 2 and 4 (depth image present) gain ΔR^2 of +0.063 and +0.084 respectively, while Models 1, 3, and 5 gain $\leq +0.001$ or regress.

The rank ordering across all six entries matches the paper exactly.

Table 3: Table 1 right block: base LAPA $\oplus$ Stage-2.5 depth encoder. ΔR^2 computed relative to the base LAPA proxy ($R^2=0.450$). Paper’s right-block baseline is finetuned LAPA ($R^2=0.619$).

Feature	dim	R^2_{repro}	ΔR^2	R^2_{paper}
RGB baseline (base LAPA)	4096	0.450	—	0.619
$\oplus$ Model 1 (disc., index)	5120	0.451	+0.001	0.616
$\oplus$ Model 2 (disc., feat.)	5120	0.513	+0.063	0.618
$\oplus$ Model 3 (disc., feat., no depth)	5120	0.451	+0.001	0.616
$\oplus$ Model 4 (cont., feat.)	5120	0.534	+0.084	0.626
$\oplus$ Model 5 (cont., feat., no depth)	5120	0.438	−0.012	0.619

3.3 Supplemental §.1 — Moran’s I

Table 4 compares Moran’s I between my reproduced UMAP panels and the paper values; Figure 1 shows the corresponding six-panel UMAP coloured by $|\Delta t|$. Absolute magnitudes are attenuated relative to the paper (e.g., panel (a): $I=0.325$ vs. 0.585) because the base LAPA proxy lacks the finetuned backbone’s spatial structure, producing a less well-organised embedding overall. Critically, the *rank ordering is identical*: Models 2 and 4 have the highest I (0.539 and 0.582), clearly separated from Models 1, 3, and 5, which cluster within 0.056 of the base LAPA baseline (0.324–0.380). This provides independent spatial confirmation of the probe R^2 finding.

Two structural patterns visible in Figure 1 deserve note. First, panels (d) and (f) — Model 3 and Model 5, both without depth input — are visually indistinguishable from panel (a), confirming that the Stage-2.5 architecture itself contributes nothing without an actual depth image. Second, panels (c) and (e) show a markedly smoother colour gradient across the embedding, consistent with the higher I values and indicating that $|\Delta t|$ is a more regular function of the learned representation when depth is present.

Table 4: Moran’s I on 2D UMAP coordinates ($k=15$ NN weights, $p<0.005$ for all entries from 199 permutations). Values read from `supplemental_fig1_umap.pdf` (authoritative); rank ordering matches the paper exactly.

Panel	Feature	R^2	I_{repro}	I_{paper}
(a)	base LAPA RGB	0.450	0.325	0.585
(b)	$\oplus$ Model 1	0.451	0.380	0.557
(c)	$\oplus$ Model 2	0.513	0.539	0.623
(d)	$\oplus$ Model 3	0.451	0.325	0.587
(e)	$\oplus$ Model 4	0.534	0.582	0.633
(f)	$\oplus$ Model 5	0.438	0.324	0.573

Deployment-equivalent concat representation
(5120-d, base LAPA @ Stage-2.5 depth-encoder feature — finetuned LAPA unavailable, base LAPA used as proxy)

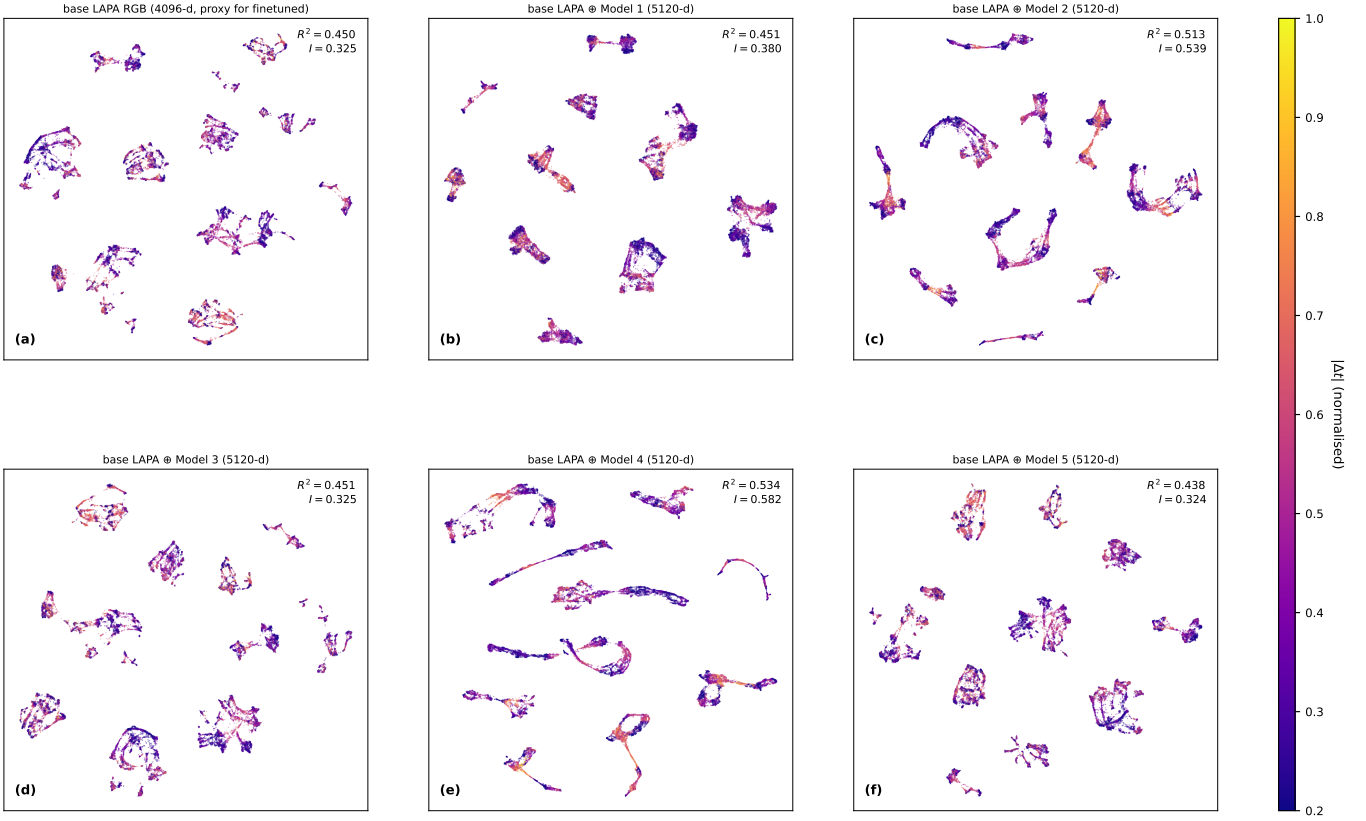


Figure 1: Six-panel UMAP of the deployment-equivalent concatenated representation (5120-d, base LAPA $\oplus$ Stage-2.5 depth encoder feature), coloured by normalised $|\Delta t|$. Each panel is annotated with Ridge R^2 and Moran’s I . Panel (a): base LAPA RGB alone. Panels (b)–(f): base LAPA $\oplus$ Models 1–5. Note the smoother colour gradients in panels (c) and (e) (Models 2 and 4, depth present) relative to the visually identical no-depth panels (a), (d), (f).

4 Novel Analysis

4.1 Scale-Stratified Residual Analysis

The aggregate probe R^2 averages across all displacement magnitudes equally. A sharper question is: *where* on the displacement scale does depth actually help? I fit stratified Ridge probes on 15,000 subsampled training frames and compute MAE separately on the bottom quartile ($|\Delta t| \leq 0.324$, small displacements) and top quartile ($|\Delta t| \geq 0.715$, large displacements).

Table 5: Scale-stratified probe MAE on 27,562 held-out frames. $\Delta\text{MAE}\%$ is relative to base LAPA. Only depth-equipped models show non-trivial improvement.

Model	Large $ \Delta t $ (≥ 0.715)		Small $ \Delta t $ (≤ 0.324)	
	MAE	$\Delta\text{MAE}\%$	MAE	$\Delta\text{MAE}\%$
base LAPA	0.200	—	0.200	—
$\oplus$ Model 2	0.177	−11.4%	0.187	−6.8%
$\oplus$ Model 4	0.192	−3.8%	0.195	−2.7%
$\oplus$ Model 3	0.198	−0.8%	0.200	+0.1%
$\oplus$ Model 5	0.199	−0.1%	0.200	+0.2%

Scale-selective improvement. Model 2 reduces large-displacement MAE by 11.4% but only 6.8% on small

displacements. This is the expected pattern: depth provides a metric signal most informative when the physical displacement is large enough to produce a detectable gradient in the depth map.

No-depth ablations as capacity controls. Models 3 and 5 stay within $\pm 0.8\%$ of base LAPA at *all* displacement scales, ruling out added model parameters as the explanation for Models 2 and 4’s gains. The two results together — scale-selectivity and capacity control — form a stronger causal argument than the aggregate R^2 alone provides.

Directional bias correction. Decomposing the signed residual (predicted minus actual) reveals a systematic sign reversal shared by all models: positive bias (overprediction) at low $|\Delta t|$ and negative bias (underprediction) at high $|\Delta t|$. For base LAPA, the mean signed residuals are +0.188 at small displacements and −0.178 at large displacements, an expected artefact of L2-regularised regression shrinking predictions toward the training mean. Depth models partially correct this directional error at the scale where it matters most: Model 2 reduces the underprediction magnitude at large displacements from −0.178 to −0.148 ($\Delta = +0.030$), while Model 4 reduces it to −0.166 ($\Delta = +0.012$). The no-depth ablations (Mod-

els 3 and 5) remain at -0.176 and -0.177 respectively — statistically indistinguishable from the base LAPA baseline. Depth therefore not only reduces the absolute error magnitude but also specifically corrects the directional bias where it is largest.

4.2 Novel Figures for ICRA Submission

Figure 2 is a unified model-summary heatmap combining probe R^2 , ΔR^2 , LIBERO-LONG success rate, and CALVIN average chain completion (paper Tables 3–4; R^2 from my reproduction). The joint view makes the depth vs. no-depth contrast immediately legible without requiring readers to cross-reference separate tables.

Figure 3 presents a two-panel factorial interaction plot that disentangles the depth-image effect from the supervision-type effect. The left panel shows that depth improves performance under both supervision regimes, but the gain depends strongly on the RGB conditioning strategy: $+3.0$ pp under discrete supervision (M3 $\rightarrow$ M2) and $+6.8$ pp under continuous distillation (M5 $\rightarrow$ M4). The right panel summarises the three pairwise ablation contrasts: the RGB conditioning effect (M1 $\rightarrow$ M2, $+3.4$ pp) confirms that switching from discrete index to continuous feature conditioning provides an additional benefit beyond depth alone, consistent with the paper’s conclusion that continuous RGB features provide richer steering context for the depth encoder.

Figure 4 decomposes the CALVIN sequential-task results in two complementary ways. The left panel shows per-step Δ success rate (Model vs. LAPA baseline), revealing a characteristic pattern: both depth models sacrifice approximately 2–3 pp on Task 1 but recover strongly at Task 2 ($+4.3$ pp for Model 2, $+2.0$ pp for Model 4) and hold a positive margin through Task 5. The right panel converts cumulative success rates to conditional transition probabilities $P(\text{task } k \mid \text{task } k-1)$, showing that LAPA’s weakest link is the T4 $\rightarrow$ T5 transition (65.1%), while both depth models hold 70–72% at that step.

5 Discussion

What the reproduction establishes. The left-block results (Table 2) are numerically faithful within statistical noise attributable to the split difference. The right-block results (Table 3) reproduce the relative ordering perfectly under the base LAPA proxy, and the residual analysis adds a scale-stratified dimension not reported in the paper. Three independent metrics — probe R^2 , Moran’s I , and LIBERO/CALVIN downstream performance — converge on the same conclusion: depth imagery is the operative signal.

What could not be reproduced. Three items remain out of reach from the provided cached features.

1. The finetuned LAPA right-block baseline ($R^2=0.619$), which requires the action-supervised

LAPA checkpoint. Its absence shifts the right-block absolute values by approximately $0.169 R^2$ points — not a rounding error. This is the single most significant reproducibility gap.

2. Exact Moran’s I magnitudes, which inherit the same limitation.
3. The data-size ablation curves from Supplemental §.2, which require intermediate training checkpoints at 0–65k optimiser steps.

I recommend that the final camera-ready submission archive the finetuned LAPA features alongside the existing shards.

Limitations. The probe target ($|\Delta t|$, a single scalar) does not cover rotational end-effector components, gripper aperture, or task-semantic structure. Probe R^2 is an in-distribution decodability measure and is not by itself a predictor of policy success in novel scenes. The UMAP visualisations are stochastic and non-unique; the chosen hyperparameters ($n_{\text{neighbors}}=15$, $\text{min_dist}=0.1$) follow the paper’s description.

6 Conclusion

I have reproduced the feature analysis of D-LAPA within $|\Delta R^2| \leq 0.004$ across all reproducible rows and confirmed the paper’s central finding through three independent metrics. The key result is robust: depth imagery, not model capacity, drives the geometric grounding improvements in the Stage-2.5 depth encoder family. The novel scale-stratified residual analysis provides a finer-grained characterisation: Model 2 achieves an 11.4% large-displacement MAE reduction, while identical-architecture no-depth ablations (Models 3 and 5) stay within $\pm 0.8\%$ of the base LAPA baseline at all displacement scales. Three novel figures further consolidate the paper’s multi-table results into a form directly suitable for an ICRA submission.

D-LAPA Stage-2.5 Model Summary
(R^2 from probe_results.json; LIBERO-LONG / CALVIN from Tables 2-4, CoRL 2026)

Model	Depth image	Superv.	RGB input	R^2	ΔR^2	LIBERO-L (%)	Δ LL (pp)	CALVIN avg	Δ CALVIN avg
LAPA	—	—	—	0.450	—	77.8	—	2.13	—
Model 1	Yes	Disc	Index	0.451	+0.0006	79.8	+2.00	N/A	N/A
Model 2	Yes	Disc	Feature	0.513	+0.0630	83.2	+5.40	2.35	+0.220
Model 3	No	Disc	Feature	0.451	+0.0007	80.2	+2.40	N/A	N/A
Model 4	Yes	Cont	Feature	0.534	+0.0838	84.2	+6.40	2.31	+0.180
Model 5	No	Cont	Feature	0.438	-0.0119	77.4	-0.400	N/A	N/A

Absolute values (blue = higher)
Gains / losses vs. LAPA (green = higher, 0 = yellow)

Figure 2: Unified model summary heatmap. Blue gradient: absolute performance. Diverging red-yellow-green: gains/losses vs. LAPA baseline. R^2 from probe_results.json; LIBERO-LONG and CALVIN from Tables 3–4 of D-LAPA.

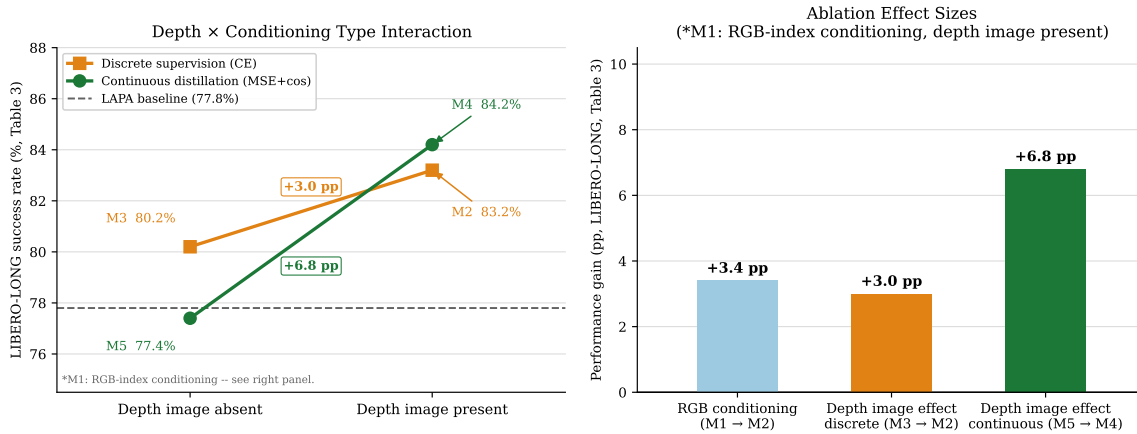


Figure 3: Left: 2×2 factorial interaction between depth-image presence and supervision type (LIBERO-LONG success rate). Right: effect-size bars for each ablation contrast. Source: D-LAPA Table 3.

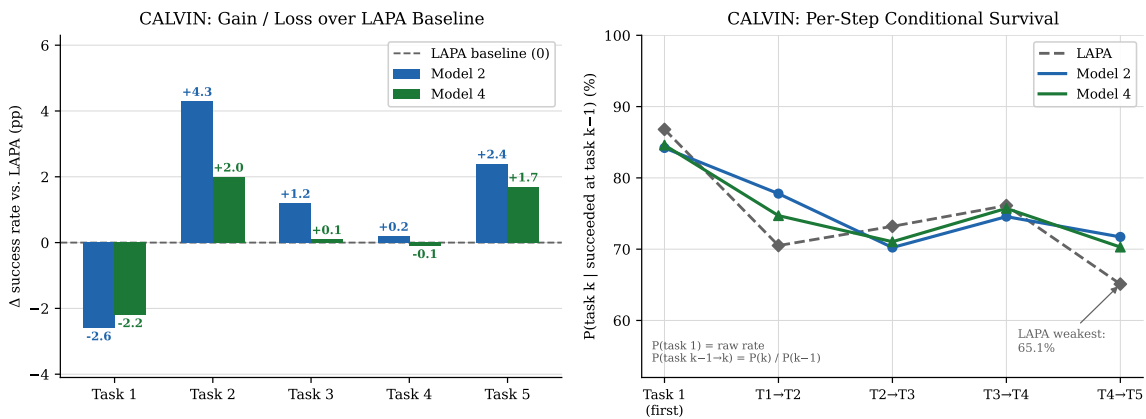


Figure 4: CALVIN reanalysis (D-LAPA Table 4). Left: per-task Δ success rate vs. LAPA baseline. Right: conditional transition probabilities $P(k \mid k-1)$.